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Subject: Fundamentals of Comptuing and programing
Assignment #1
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1 Theorical questions

1. Explain the characteristics and importance of algorithms.

   (a) Finiteness: The algorithm must always terminate after a finite number of steps. It cannot loop infinitely.
   (b) Definiteness: Each step must be rigorously and unambiguously defined. There is no room for interpretation (e.g., "add a small number" is invalid; "add 1" is definite).
   (c) Input: It takes zero or more defined inputs (initial quantities provided before execution begins).
   (d) Output: It produces at least one output (the result that has a specified relation to the inputs).
   (e) Effectiveness: Every operation must be basic enough that it could, in principle, be carried out exactly and in a finite amount of time using just pen and paper.

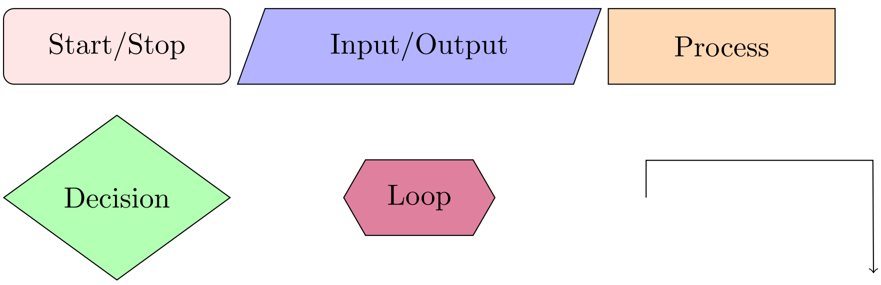
   Algorithms are important as they provide a standardized, language-agnostic way to solve problems efficiently.

2. Distinguish between algorithms, flowcharts, pseudocode, and programs.

   (a) Algorithm: The abstract logical concept or method.
   (b) Flowchart: A text-based, structured draft of the algorithm.
   (c) Pseudocode: A graphical representation of the algorithm's topology.
   (d) Programs: The concrete implementation written for a machine.

3. Discuss various symbols with labeled diagrams.

   (a) Start/Stop: Marks the absolute Start and End points of the algorithm. Every flowchart must begin and terminate with this symbol.
   (b) Input/Output: Represents data entering the system or leaving the system.
   (c) Process: Represents an operational step, such as a mathematical calculation, variable assignment, or data manipulation.
   (d) Decision: The branching point where the logic evaluates a condition. True and False
   (e) Loop (hexagon): Represents the looping condition. It could be for loop, while loop or do-while loop.
   (f) Flowlines (Arrows): The directional vectors that connect the symbols, establishing the strict chronological sequence of execution.



2 Sequence type

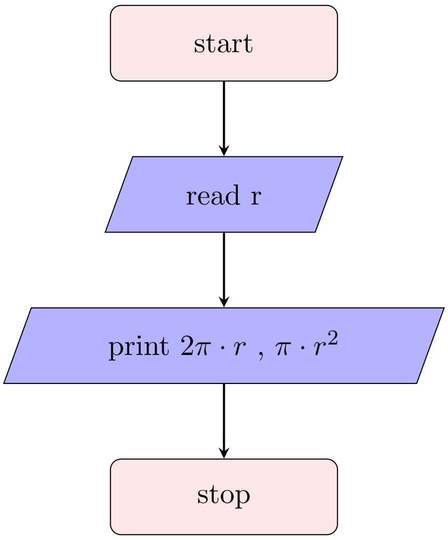
1. Write an algorithm and draw the flowchart to find the area and perimeter of the circle

   1: start

   2: read the radius r

   3: print the circumferance $2\pi r$ and area πr^2

   4: stop



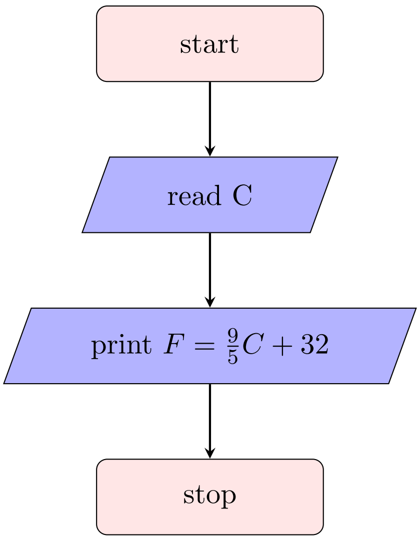
2. Write an algorithm and draw a flowchart to convert temperature from Celsius to Fahrenheit. and vice versa

   1: start

   2: read the temperature in Celsius C

   3: print $F = \frac{9}{5}C + 32$

   4: stop

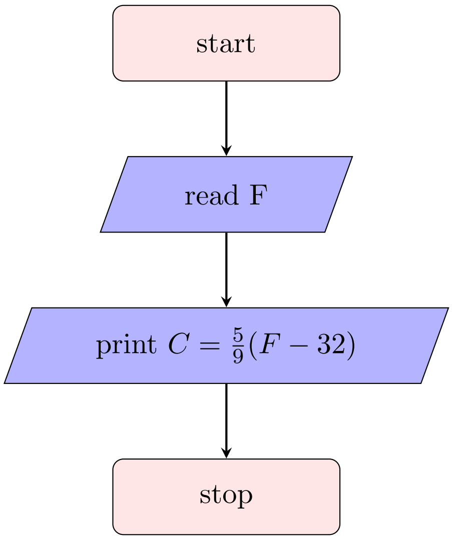


   1: start

2: read the temperature in Fahrenheit F

3: print $C = \frac{5}{9}(F - 32)$

4: stop



3. Write an algorithm and draw the flowchart to determine the Simple Interest, reading Principle, Rate and Time.

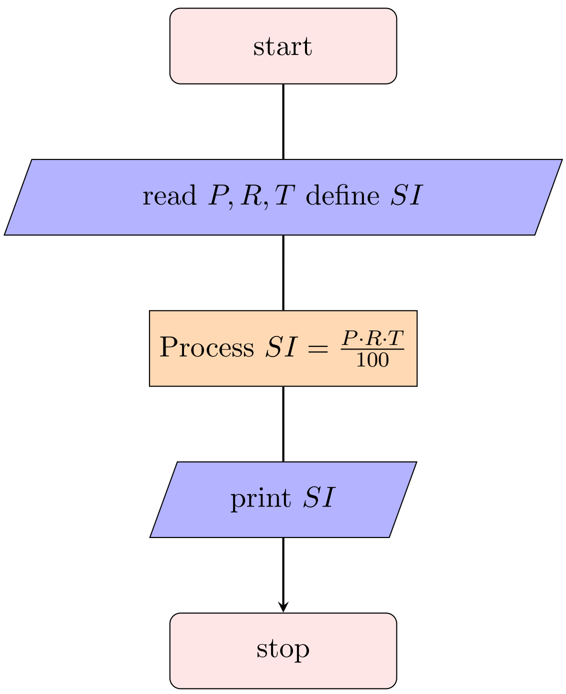
1: start

2: read the Principle P, Rate R and Time T, define the Simple Interest SI

3: Process $SI = \frac{P \cdot R \cdot T}{100}$

4: print SI

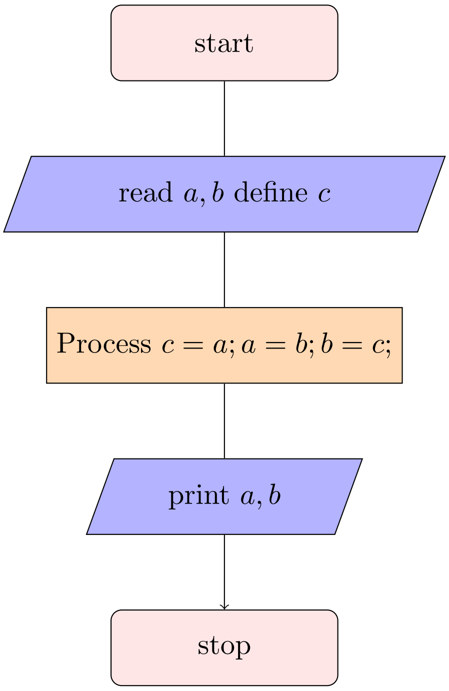
5: stop



4. Write algorithms and flowcharts:

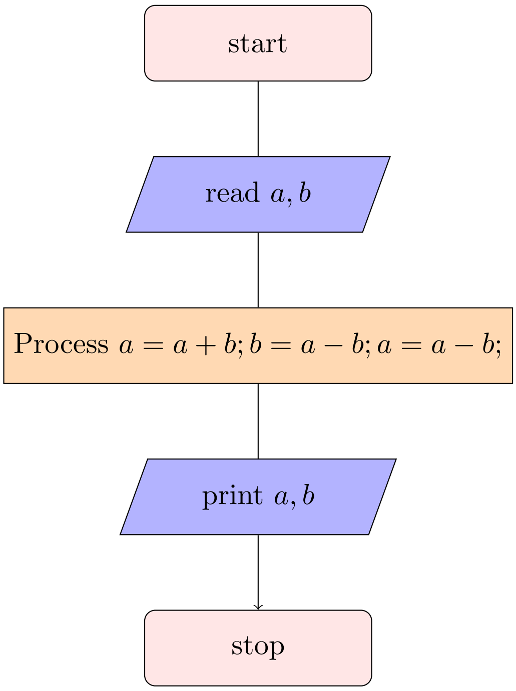
(a) Using a third variable

```
start
read a,b,c
process c=a; a=b; b=c;
print a,b
stop
```



(b) Without using a third variable

```
start
read a,b
process a=a+b;b=a-b;a=a-b;
print a,b
stop
```



3 Conditional type

1. Write an algorithm and draw the flowchart to check whether number is even or odd.

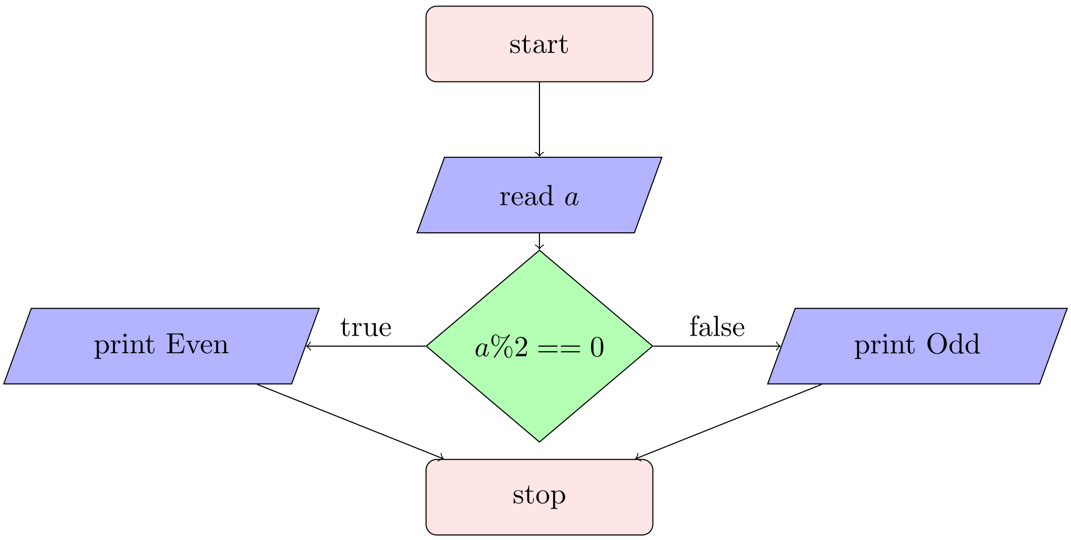
   1: start

   2: read a

   3: process if the remainder of a when divided by 2 is 0

   4: print (if true) even; (if false) odd

   5: stop



2. Write an algorithm and draw the flowchart to check whether given year is a leap year or not.

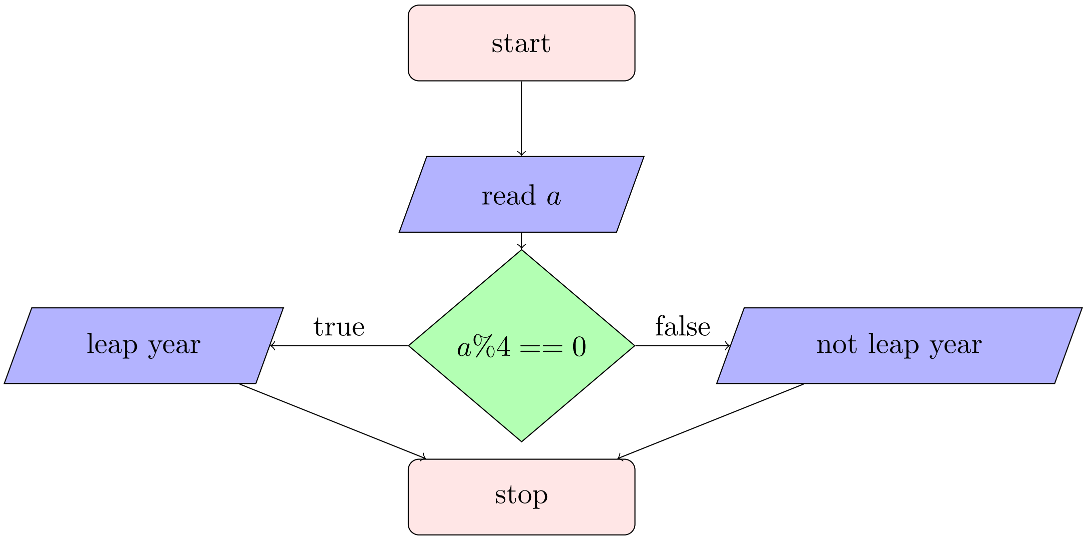
   1: start

2: read a

3: process if the remainder of a when divided by 4 is 0

4: print (if true) leap year; (if false) not

5: stop



3. Write an algorithm and draw the flowchart to check if a triangle is equilateral triangle.

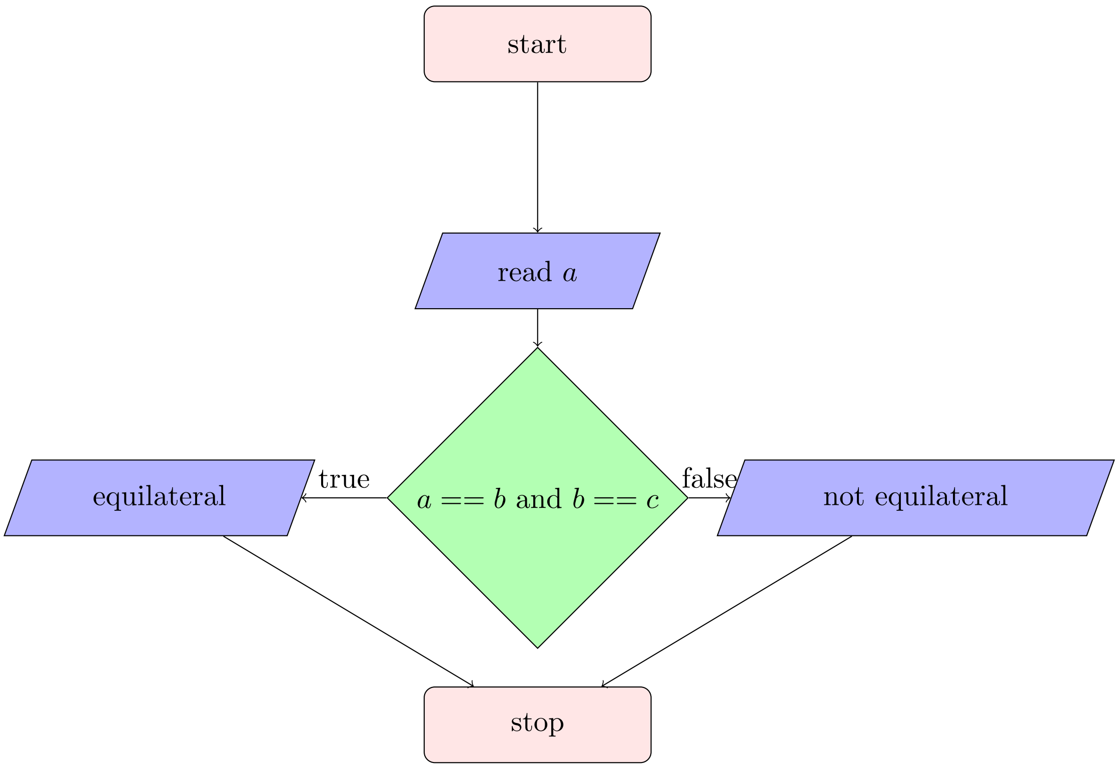
1: start

2: read a, b, c

3: process a==b==c is true or not

4: print (if true) equiv; (if false) not equilateral

5: stop



4. Write an algorithm and draw the flowchart to find largest of three numbers.

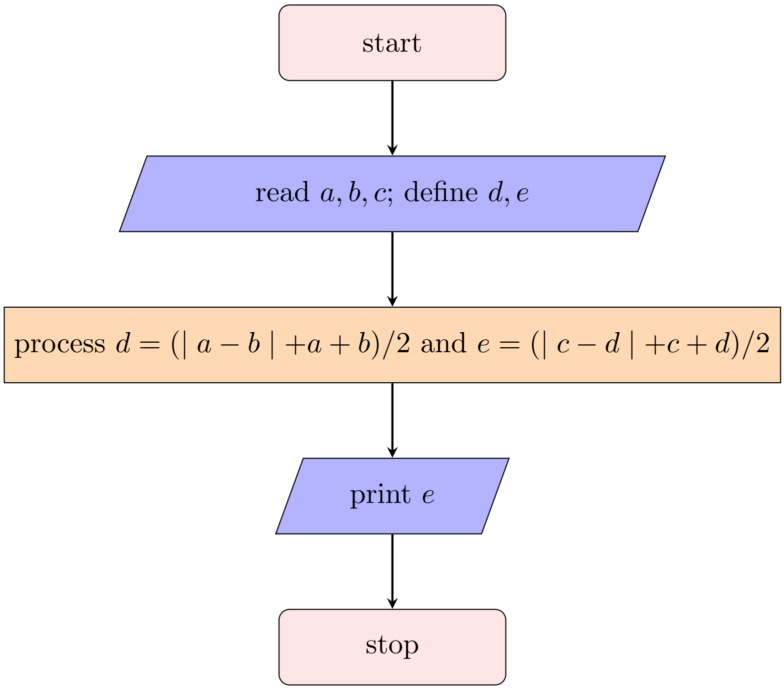
   1: start

   2: read a, b, c and define m, e

   3: process $d = (| a - b | + a + b)/2$ and then process $e = (| c - d | + c + d)/2$

   4: print e

   5: stop



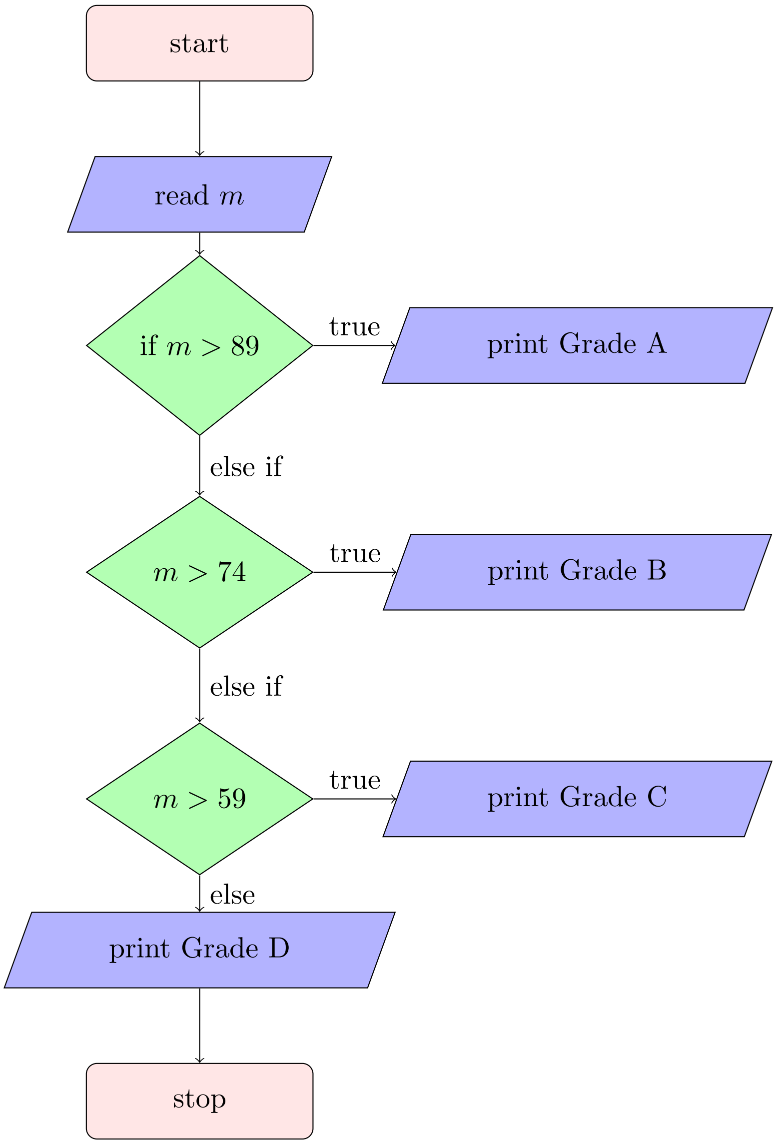
5. Write an algorithm and draw the flowchart to read the marks (0-100) and display the grade. (Grades: A(90-100), B(75-89), C(60-74), F(Below 60).

   1: start

   2: read m

   3: (if $m > 89$) print Grade A; (if $m > 74$) print Grade B;(if $m > 59$) print Grade C;(else) print Grade D

   4: stop



4 Repetition type

1. Write an algorithm and draw the flowchart to find factorial of a number

   1: start

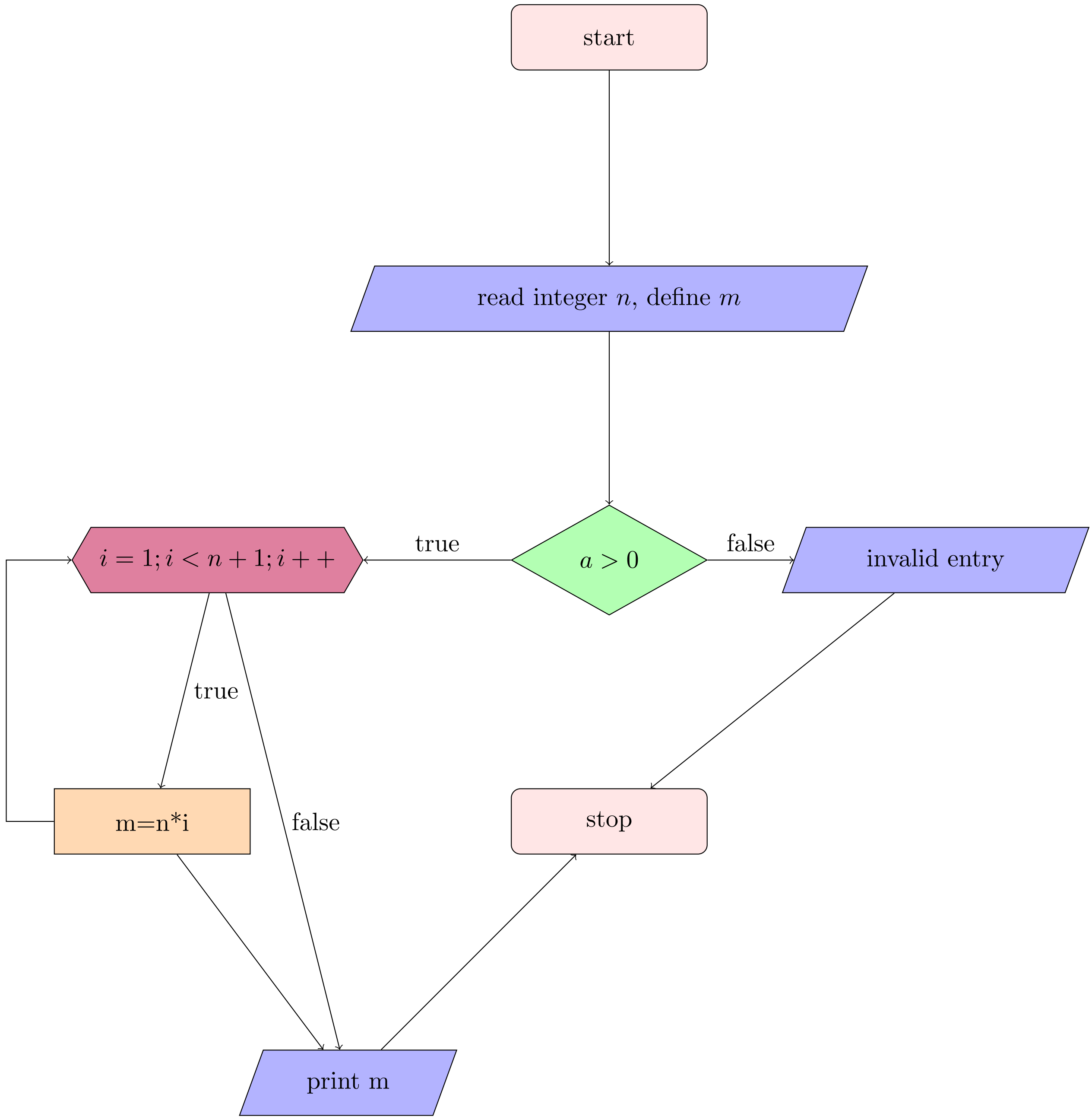
   2: read n define m

   3: check if the number is positive integer or not

   4: (if true) process $m = n \cdot (n-1) \cdots 2 \cdot 1$; (if false) invalid entry

   5: print m

   6: stop



2. Write an algorithm and draw the flowchart to check if a number is palindrome or not.

3. Write an algorithm and draw the flowchart to find sum of digits of a given number.

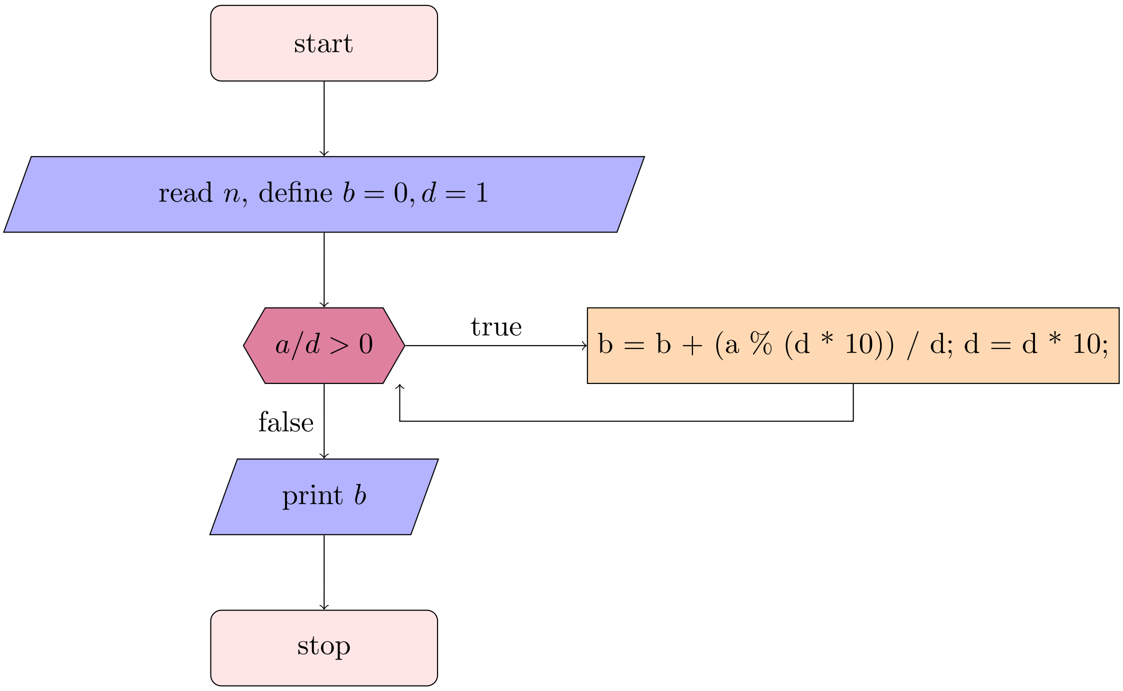
   1: start

   2: read n, define $b = 0, d = 1$

   3: check if $a/d > 0$ or not

   4: (if true) process $b = b + (a\%(d * 10))/d$ and $d = d * 10$ (if false) print b

   5: stop



4. Write an algorithm and draw the flowchart to check if the entered phone number is 10 digit

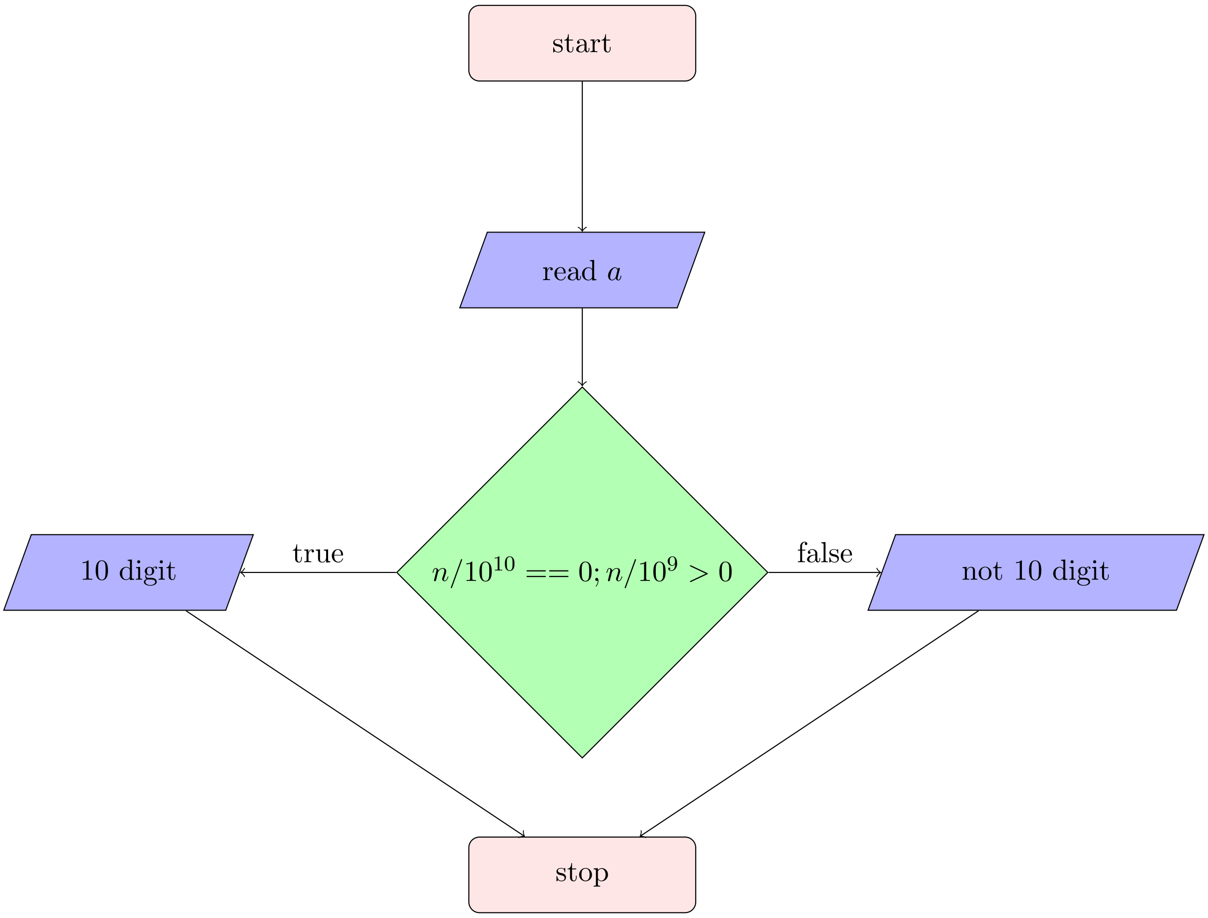
   1: start

   2: read int n

   3: process if $n/10^{10}$ is 0 and $n/10^9 \geq 1$

   4: print (if true) yes 10 digit; (if false) not 10 digit

   5: stop



5. Write an algorithm and draw the flowchart to find sum of first N natural numbers.

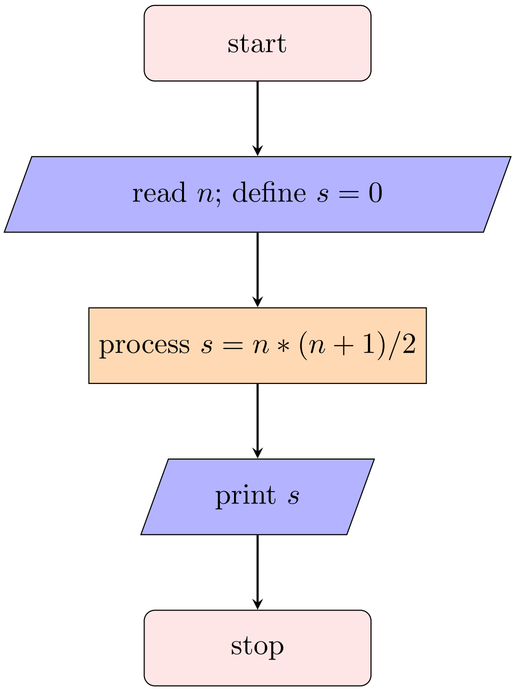
    1: start

    2: read $n, s = 0$

    3: process $s = \frac{n(n+1)}{2}$

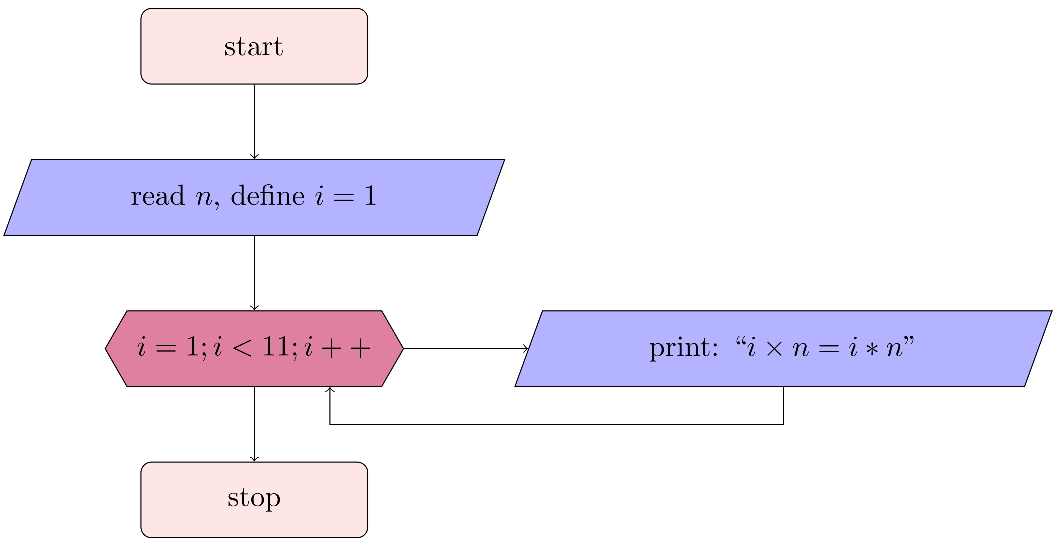
    4: print s

    5: stop



6. Write an algorithm and draw the flowchart to Generate the multiplication table for a given number up to 10.

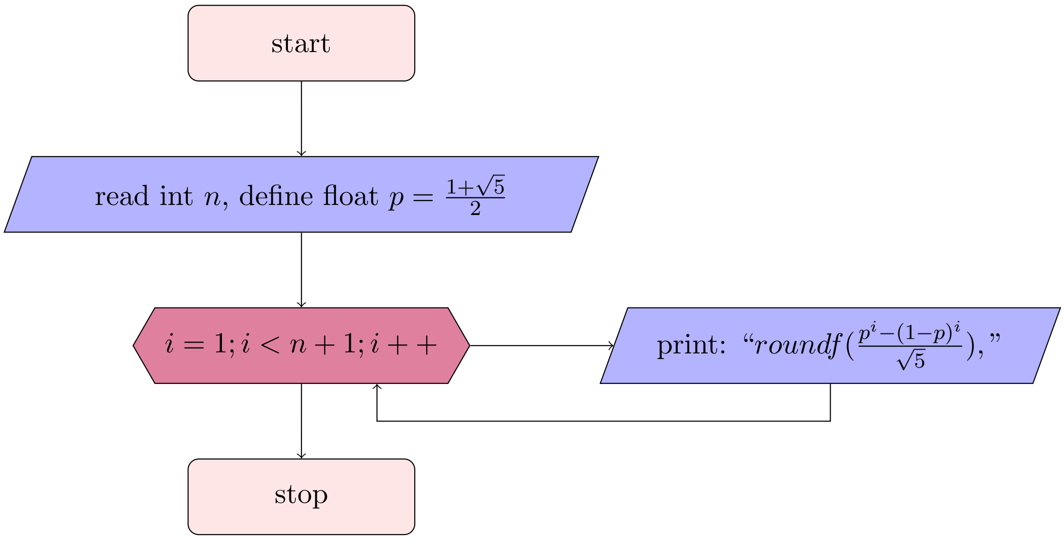
    1: start
    2: read n define $i = 1$
    3: print "$i \times n$" $= i \cdot n$ for $1 \leq i \leq 10$
    4: stop



7. Write an algorithm and draw the flowchart to Generate the first N terms of the Fibonacci series.

    1: start
    2: read integer n and float $p = \frac{1+\sqrt{5}}{2}$
    3: print closest integer value of $\frac{p^i-(1-p)^i}{\sqrt{5}}$ for $1 \leq i \leq n$

4: stop

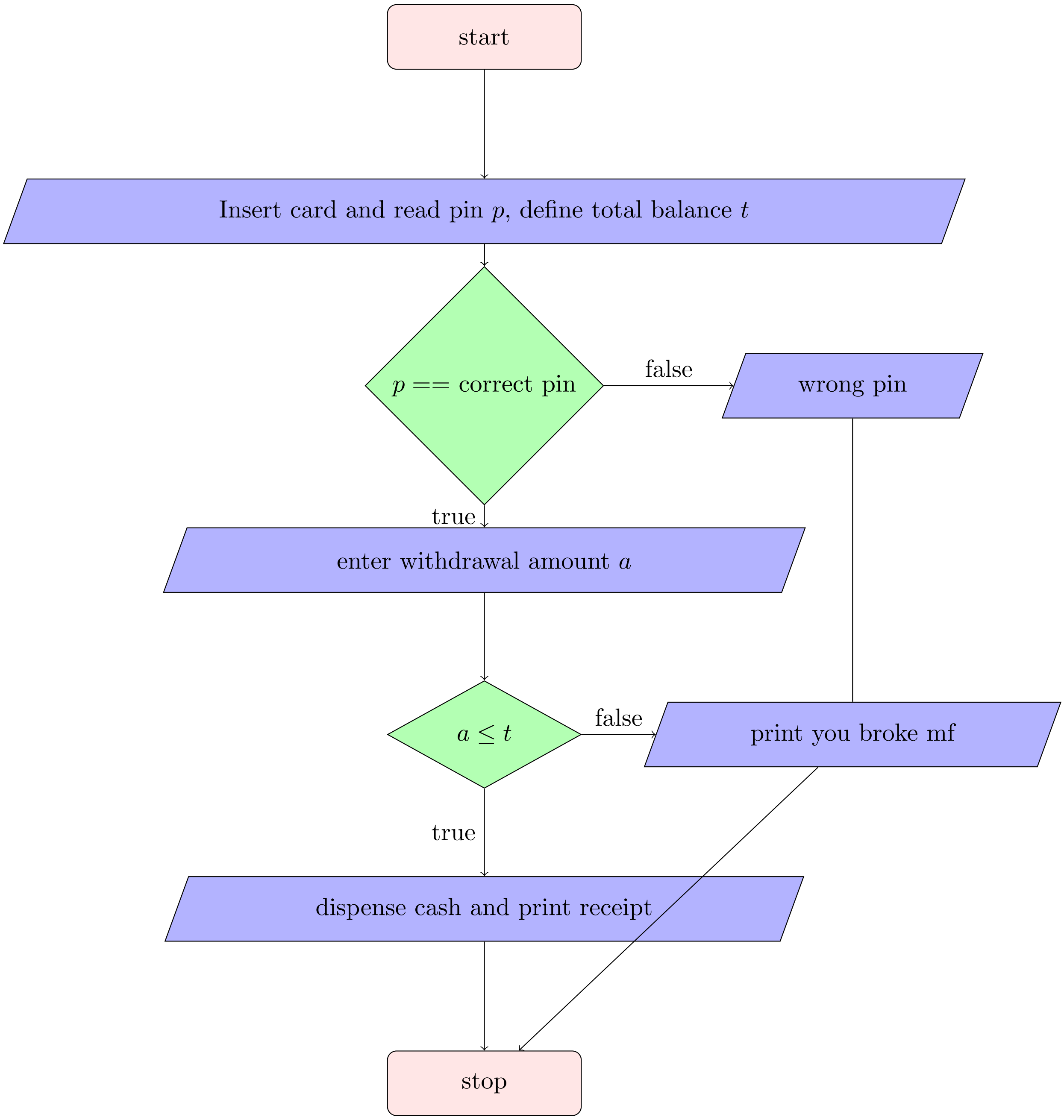


5 Using all

1. Design an algorithm and flowchart for an ATM system. Include appropriate decision blocks. Steps should include:

    i. Insert Card
    ii. Enter PIN
    iii. Validate PIN
    iv. Select Withdrawal
    v. Enter Amount
    vi. Check Balance
    vii. Dispense Cash
    viii. Print Receipt
    ix. Exit

1: start
2: Insert card, read pin p
3: pin validate (if true) go to next step (if false) print wrong pin
4: select withdrawal and enter amount
5: check balance
6: sufficient balance (if true) dispense cash (if false) print you broke mf
7: print receipt
8: stop



2. Design an algorithm and flowchart to:
   Read marks in five subjects, Calculate total, Calculate average , Determine grade , Display Pass/Fail
   Rules: Pass if every subject has marks ≥ 40,
   Grade: A (≥ 90) , B (75-89) , C (60-74) , D (40-59) , F (< 40)

   1: start

   2: read m_1, m_2, m_3, m_4, m_5 define $t, a,$

   3: process $t = m_1 + m_2 + m_3 + m_4 + m_5$ and $a = t/5$

   4: (if $m_1 < 40$ or $m_2 < 40$ or $m_3 < 40$ or $m_4 < 40$ or $m_5 < 40$) print FAIL with grade F
   (else if $a \geq 90$) print PASS with grade A
   (else if $a < 90$ && $a \geq 75$) print PASS with grade B

(else if $a < 75$ && $a \geq 60$) print PASS with grade C
(else if $a < 60$ && $a \geq 39$) print PASS with grade D

5: stop